ORIGINAL ARTICLE

Systems engineering; model-based systems engineering; nanosatellites; electrical power systems

Model-Based Systems Engineering for Nanosatellite Electrical Power-System Development: Systematic Review of Method, Evidence, and Pre-Physical Verification

Author Information To Be Supplied[1]

1 | METHODOLOGY

1.1 | Protocol and research questions

This systematic literature review followed the protocol maintained for SLR1. The protocol defines a nanosatellite-class population, an MBSE intervention, and outcomes concerned with general modeling, EPS-relevant value, and verification before physical development. The review method is intentionally narrower than broad space-MBSE syntheses: Buruso et al. Buruso et al. (2026) are used as a methodological and positioning comparator, especially for transparent review procedure and concise enabler classification, but their review is not treated as evidence about the papers in this corpus.

The protocol research questions were:

RQ1. How can nanosatellite missions and EPS be modeled in a general manner with minimal knowledge about their behavior?

RQ2. What are the claimed outcomes related to reliability, reproducibility, and integrability derived from EPS modeling in nanosatellite missions?

RQ3. How have verification procedures been integrated into EPS models prior to physical development?

The population was interpreted as CubeSat and nanosatellite-class missions rather than larger spacecraft or smaller educational platforms such as CanSats. For RQ1, the focal intervention was system-level or EPS-level MBSE. For RQ2, outcomes had to be attributable to MBSE artifacts, MBSE activities, or model-enabled process changes rather than to the selected EPS technology alone. For RQ3, the review looked for requirements, verification, validation, IVV, test, simulation, or analysis artifacts integrated in or around the model before hardware development.

1.2 | Information sources and search strategy

The database scope was Scopus and IEEE Xplore. Scopus was searched directly. IEEE Xplore coverage was incorporated through manual CSV exports that were screened under the same protocol. Google Scholar exact-title fallback was used only during the later citation-chasing continuation to confirm or recover specific citation-chasing targets after Scopus misses; it was not used as an open-ended third database.

Search strings combined nanosatellite/CubeSat population terms with MBSE terms such as “model-based sys-

spacecraft, payload-only, ground-segment-only, software-only, hardware-calibration, education/tutorial, proceedings-volume, and non-space records unless screening recorded a specific protocol-based rationale for retention;
- include studies using MBSE, SysML, model-based architecture, model-based requirements, model-based simulation/analysis, or closely related digital-engineering approaches at system level, mission level, EPS level, or an EPS-adjacent power/electrical-architecture level;
- for RQ1, prioritize system-level or EPS-level models, reference models, templates, reusable workflows, behavior models, and architecture models over isolated subsystem models;
- for RQ2, retain claimed outcomes only when the outcome could be linked to MBSE artifacts or model-enabled processes, not merely to EPS component choices;
- for RQ3, require explicit verification, validation, IVV, test, requirements traceability, simulation, parametric checking, or pre-physical-development evaluation evidence in or around the model.

Screening distinguished core and supporting studies. Core papers directly shaped the synthesis because they were central to CubeSat/nanosatellite MBSE, EPS/power modeling, traceability, reusable reference models, or V&V. Supporting papers were retained when they provided contextual evidence on toolchains, lifecycle continuity, requirements-in-the-loop, risk, project-management, or related nanosatellite MBSE practices without being as central to the protocol RQs.

1.4 | Data extraction and quality appraisal

Extraction was performed on the 40 accessible reports, comprising 21 core and 19 supporting studies. The four reports that remained inaccessible at retrieval were retained in corpus-management artifacts but were not included in the extraction synthesis.

A standardized extraction schema was used for every accessible study. Extracted fields covered bibliographic and review-management metadata; research-question relevance; mission/platform context; lifecycle stage; EPS relevance; model maturity; MBSE objective, scope, artifacts, abstraction level, and reuse mechanism; implementation pathway and tailoring; toolchain, interoperability, and lifecycle-continuity mechanisms; EPS and verification-specific artifacts; claimed outcomes, mechanisms, evidence types, metrics, baselines, and confidence ratings; governance, ROI, deliverables, and project-decision linkage; gap mapping; limitations; direct evidence quotes; lineage notes; and extraction confidence.

Quality appraisal used the protocol checklist rather than an exclusion threshold. Each checklist item was scored as No = 0, Insufficiently = 0.5, or Yes = 1. The checklist assessed venue credibility; whether limitations, threats, assumptions, or scope boundaries were discussed; clarity of nanosatellite/CubeSat context and lifecycle stage; clarity of MBSE method, language, toolchain, model artifacts, and system boundary; whether claimed outcomes were explicitly linked to MBSE artifacts or activities; whether evidence was provided for claims; whether requirements, verification, validation, design decisions, or lifecycle gates were traced to model artifacts when relevant; and whether model availability, reusable assets, data exchange, or implementation details were reported. Quality scores were used to inform sensitivity, confidence, and discussion of evidence strength; they were not used to automatically remove studies after full-text extraction.

No independent inter-rater reliability exercise is documented in the review artifacts. Consequently, the review does not report agreement statistics. Instead, extraction consistency was supported by the fixed protocol, a common extraction schema, explicit rationales and direct quotes for coding decisions, extraction-confidence ratings, and cross-checks against the corpus, screening, retrieval, and prior scoping-synthesis records.

1.5 | Synthesis strategy

The synthesis was structured by the three protocol RQs rather than by a general MBSE tool or method taxonomy. Descriptive statistics were used to summarize corpus characteristics, study tiers, EPS relevance, approach classifications, quality-score distribution, research-question relevance, value-claim evidence, and V&V linkage categories. Narrative synthesis then interpreted these counts in relation to the protocol questions.

A concise enabler taxonomy was retained only as a secondary explanatory layer. The Pure, Enhanced, Extended, and Integrated MBSE/digital-engineering categories, together with lifecycle activity anchors such as requirements management, model development, simulation/analysis, configuration management, and collaboration/communication, were used as sensitizing categories consistent with the broader space-MBSE review style of Buruso et al. Buruso et al. (2026). The main synthesis, however, focused on two issues specific to this review: the evidence behind claimed value and the linkage between MBSE artifacts and V&V before physical development.

For RQ1, studies were compared by modeling focus: whole-nanosatellite models that include EPS or power as a subsystem, generic whole-nanosatellite models without explicit EPS, and EPS-only or EPS-first models. Reuse mechanisms such as reference models, templates, profiles, libraries, open packages, and repeatable workflows were examined as evidence for generality and for modeling with limited prior behavioral knowledge. For RQ2, each value claim was synthesized by outcome domain, specificity, MBSE mechanism, evidence type, metric or value indicator, baseline or comparator, and evidence-confidence rating. This prevented asserted benefits such as reliability, reproducibility, integrability, schedule reduction, or communication improvement from being treated as measured effects unless the extraction recorded supporting evidence. For RQ3, V&V linkage was synthesized across requirements-to-model traceability, verification matrices, parametric and power-budget checks, executable simulation or scenario analysis, test cases and procedures, automated tests, external tool/data-flow coupling, and evidence of physical, software-in-the-loop, hardware-in-the-loop, or integration-test readiness.

1.6 | Review validity

Several measures were used to strengthen review validity while preserving the limits of the available artifacts. Construct validity was supported by deriving eligibility, extraction fields, and synthesis categories from the protocol RQs and by keeping the broad space-MBSE review Buruso et al. (2026) as a positioning reference rather than as corpus evidence. Search validity was supported by using two major engineering bibliographic sources, exact query logging, deduplication, targeted recall searches for EPS/V&V sparsity, IEEE parity checks, and citation-chasing recovery for specific missing titles. The main search limitation is that the database scope was restricted to Scopus and IEEE Xplore, with Google Scholar used only for exact-title fallback, so grey literature and some publisher-specific records may be underrepresented.

Selection validity was supported by explicit inclusion/exclusion criteria and a maintained screening log. Residual uncertainty remains for records with limited abstracts, documented borderline/contextual inclusions, and inaccessible reports. Extraction validity was supported by full-text extraction for accessible studies, direct evidence quotes, quality appraisal, claim-evidence coding, and extraction-confidence ratings. The absence of a documented inter-rater exercise is a limitation; no reliability statistic is claimed.

Synthesis validity was addressed by separating descriptive corpus patterns from evidence-backed findings. Repeated publications from the same CubeSat reference-model lineage were treated with lineage notes so that publication count would not be mistaken for independent replication. Claims about MBSE value were interpreted through evidence confidence and baseline availability, and EPS-specific conclusions were kept conservative because the cor-

pus contains many system-level or EPS-adjacent studies but few direct EPS-first studies. External validity is therefore limited to nanosatellite/CubeSat-class MBSE under the search scope and should not be generalized to all space MBSE practice or to all spacecraft EPS development without further evidence.

2 | RESULTS

2.1 | Search, screening, retrieval, and extraction flow

The final synthesis used the 40 accessible studies for which standardized extraction files were available. The review flow was reconstructed from the search, screening, retrieval, and extraction records: 199 raw merged search records were identified, 118 duplicates were removed, 81 unique records were screened, 37 were excluded during screening, 44 reports were sought for retrieval, 4 reports were not retrieved or remained inaccessible, and 40 reports were extracted. The inaccessible retained records were kept in review-management artifacts but were not included in the evidence synthesis.

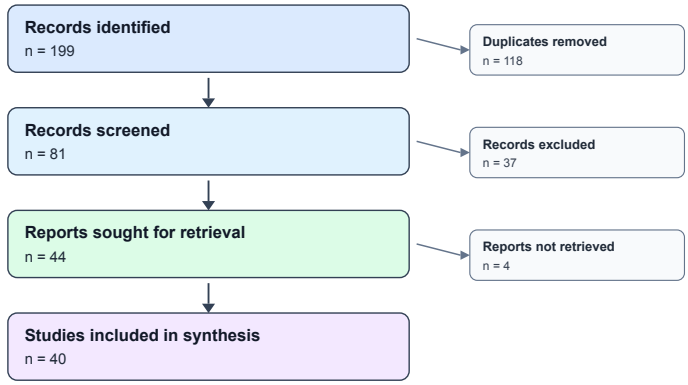


FIGURE 1 Flow of record identification, screening, retrieval, and inclusion for the review synthesis.

The extracted set comprised 21 core studies and 19 supporting studies published from 2013 to 2025. Counts below therefore refer to the 40 extracted studies unless identified as multi-label claim-domain or V&V-linkage counts.

2.2 | Study characteristics and quality

The quality checklist scores ranged from 5.0 to 7.5 on the eight-item scale, with a mean of 6.29. The score distribution was 5.0 ($n = 2$), 5.5 ($n = 7$), 6.0 ($n = 8$), 6.5 ($n = 14$), 7.0 ($n = 7$), and 7.5 ($n = 2$). Extraction confidence was coded as high for 26 studies and medium for 14 studies. RQ relevance was uneven: RQ1 was high for 11 studies and partial for 29; RQ2 was partial for all 40 studies; and RQ3 was high for 4, partial for 28, and not relevant for 8. The all-partial RQ2 pattern reflects that most papers stated MBSE value claims but did not directly measure EPS reliability, reproducibility, or integrability outcomes.

The scoping-review limitation coding further contextualizes quality: 31 studies were coded as prototype, incom-

TABLE 1 Study, quality, and claim-evidence summary.

Metric	n
Extracted studies	40
Core studies	21
Supporting studies	19
Publication year range	2013–2025
Claim records	294
Unsupported claims	45
Weak claims	152
Moderate claims	97
Strong claims	0
High-confidence extractions	26
Medium-confidence extractions	14

plete maturity, or future work; 24 as limited-scope, single-case, or partial subsystem/mode coverage; 24 as lacking quantitative evaluation, having weak validation, or having no threats discussion; 18 with data/model-quality/source/assumption limitations; and 16 with tool dependency, interoperability, licensing, or learning-curve limitations. These limitations were treated as evidence-strength constraints rather than as exclusion criteria.

2.3 | Modeling focus and EPS relevance

The current scoping-review focus classification distinguished three modeling-focus groups: 32 studies modeled a whole nanosatellite with EPS or power as a subsystem, 7 modeled a whole nanosatellite or generic CubeSat without explicit EPS modeling, and 1 study was EPS-only or EPS-first. The detailed EPS-relevance coding used for this SLR is narrower: 23 studies were coded as power-budget/electrical-architecture relevant, 8 as EPS-adjacent, 7 as system-level only, 1 as EPS-specific, and 1 as not EPS-relevant.

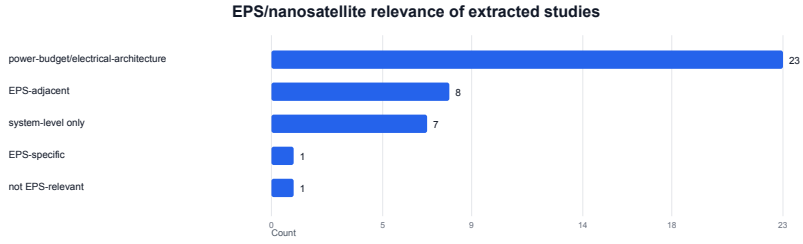


FIGURE 2 EPS relevance counts in the extracted corpus.

The only EPS-specific study was the open-source Papyrus/SysML power-subsystem model with Java-based pa-

TABLE 2 Detailed EPS relevance counts.

EPS relevance category	n
Power-budget/electrical-architecture	23
EPS-adjacent	8
System-level only	7
EPS-specific	1
Not EPS-relevant	1

rameter and requirement-traceability extraction Strobbe et al. (2023). Most EPS evidence was instead embedded in system-level models or preliminary design studies. Examples include RAX/CubeSat energy and data simulations Kaslow et al. (2014); Spangelo et al. (2013); Anderson et al. (2014), SysML parametric mass and power-margin checks Waseem and Sadiq (2018), Capella/Arcadia preliminary EPS allocation for 3ColStar Buitrago-Leiva et al. (2024), CNES concurrent electrical-architecture simulations Boan (2025), and PLM/Dymola requirements-in-the-loop power simulations Rehman and Fortin (2024). The not-EPS-relevant record, EIRSAT-1, still contributed V&V evidence through a communications-subsystem verification demonstrator rather than an EPS model Akarapu et al. (2023).

The scoping-review scope tags were broader than the SLR EPS-relevance categories: 33 studies had an EPS/power scope tag, 32 had verification in scope, 27 had budget in scope, 27 had simulation in scope, and 26 used a reference model or template. Thus, EPS appeared frequently as a system concern, but direct EPS-first MBSE evidence was sparse.

2.4 | MBSE approaches, languages, and toolchains

Approach classification was dominated by integrated or enhanced MBSE. Nineteen studies were coded as Integrated MBSE/digital engineering, 13 as Enhanced MBSE, 6 as Pure MBSE, and 2 as Extended MBSE. This distribution situates the corpus but does not by itself establish value or V&V effectiveness.

TABLE 3 MBSE approach-classification counts.

Approach classification	n
Integrated MBSE/digital engineering	19
Enhanced MBSE	13
Pure MBSE	6
Extended MBSE	2

The scoping-review tool/language counts show a SysML-centered corpus: SysML appeared in 28 studies, Cameo/MagicDraw/Magic in 16, MATLAB/Simulink in 12, STK/GMAT/DRAMA orbit tools in 9, PLM/Teamcenter/Windchill/3DExperience in 9, OOSEM in 7, ModelCenter/MBSE Analyzer in 6, spreadsheets in 5, ParaMagic/ParaSolver/acausal parametrics in 5, and Capella/Arcadia in 4. Less frequent but results-relevant toolchains included Papyrus/Eclipse for the EPS-specific open-source case Strobbe et al. (2023), Gaphor/UML stereotypes for open-source preliminary-design

formalization Luccisano et al. (2024), docs-as-code/M30ML for SeaLion documentation and traceability Chiu et al. (2023), Dymola/Modelica in 3DExperience Rehman and Fortin (2024), and IDM-CIC/Simu-CIC/Opalis for electrical-architecture concurrent engineering Boan (2025).

Toolchain use was usually reported as a means to connect requirements, architecture, analysis, and verification artifacts. The strongest toolchain evidence came from executable or data-flow demonstrations, such as MagicDraw/Cameo–ModelCenter–STK–MATLAB integrations for RAX-style CubeSat analysis Kaslow et al. (2014); Spangelo et al. (2013), Capella/ForPlan or Capella/open-source simulation coupling de Almeida et al. (2021); Campioli et al. (2025), and SysML/PLM/Dymola requirements-in-the-loop simulation Rehman and Fortin (2024). Pure reference-model studies reported reusable structures and V&V packages, but generally provided less executed analysis evidence Kaslow and Madni (2017); Kaslow et al. (2019).

2.5 | Claimed outcomes and value-evidence strength

The extraction recorded 294 claim records. Evidence confidence for those claims was coded as Unsupported (n = 45), Weak (n = 152), Moderate (n = 97), and Strong (n = 0). The generated "top value claims" artifact therefore contains 97 moderate claims and no strong claims. Outcome domains are multi-label; the 294 claims produced 907 claim-domain memberships.

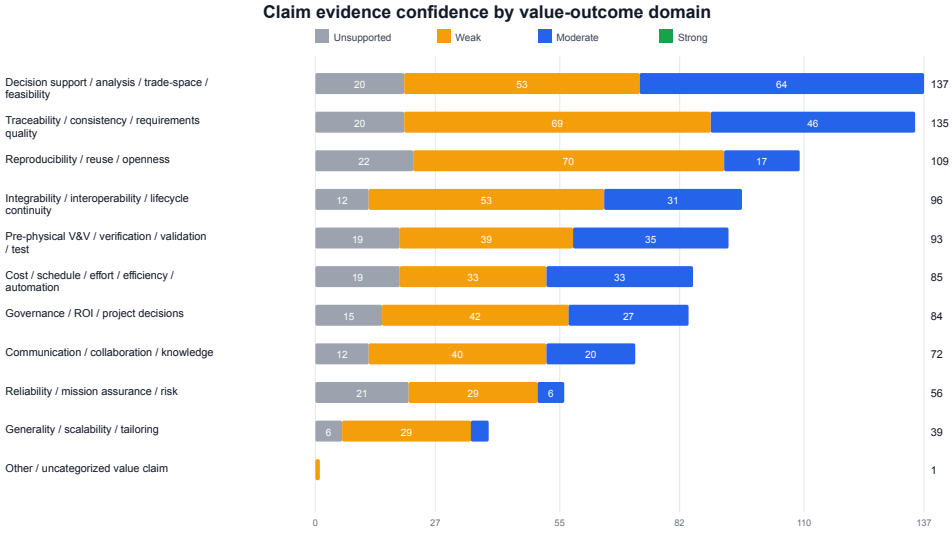


FIGURE 3 Claim evidence by value-outcome domain. Domains are multi-label, so counts exceed the number of unique claim records.

The most frequent domains were decision support/analysis/trade-space/feasibility (137 memberships across 38 studies), traceability/consistency/requirements quality (135 across all 40 studies), reproducibility/reuse/openness (109 across 36), integrability/interoperability/lifecycle continuity (96 across 38), and pre-physical V&V/test (93 across 36). EPS-specific or power-relevant claims were concentrated in decision support (29 memberships), traceability/requirements quality (13), pre-physical V&V (12), and governance/project decisions (10), while reliability/mission-assurance/risk had

TABLE 4 Claim evidence by value-outcome domain. Domains are multi-label.

Outcome domain	Total	Unsup.	Weak	Mod.	Strong	Studies
Decision support / analysis / trade-space / feasibility	137	20	53	64	0	38
Traceability / consistency / requirements quality	135	20	69	46	0	40
Reproducibility / reuse / openness	109	22	70	17	0	36
Integrability / interoperability / lifecycle continuity	96	12	53	31	0	38
Pre-physical V&V / verification / validation / test	93	19	39	35	0	36
Cost / schedule / effort / efficiency / automation	85	19	33	33	0	37
Governance / ROI / project decisions	84	15	42	27	0	35
Communication / collaboration / knowledge	72	12	40	20	0	34
Reliability / mission assurance / risk	56	21	29	6	0	28
Generality / scalability / tailoring	39	6	29	4	0	29
Other / uncategorized value claim	1	0	1	0	0	1

only 3 EPS-specific or power-relevant claim memberships.

Moderate evidence typically came from executed simulations, model-derived calculations, traceability artifacts, or case-study outputs. Examples include the RAX model's energy/data interaction and solar-panel/battery sizing analyses Kaslow et al. (2014); Spangelo et al. (2013), Papyrus/Java power-subsystem parameter-impact and requirement-fulfillment tables Strobbe et al. (2023), power and SoC simulation against requirements in a PLM/Dymola workflow Rehman and Fortin (2024), worst-case battery survival simulation with a reported minimum state of charge in an electrical-architecture process Boan (2025), and positive mass/power margins from SysML parametrics Waseem and Sadiq (2018). Project-value metrics were rarer and case-specific, such as a one-month versus three-month feasibility-study comparison and lower estimated mission cost in one concurrent-design case Bella et al. (2024), or the reported four-month preliminary-design cycle using Arcadia and weekly sprints Buitrago-Leiva et al. (2024).

Unsupported and weak claims were common where papers asserted reduced cost, reduced schedule, improved reliability, improved learning, or greater reuse without a baseline, metric, reuse trial, stakeholder evaluation, integration test, or flight/operational evidence. For example, the EPS-specific Papyrus study provided moderate evidence for traceability and requirement-constraint checking, but its cost/time savings remained a future quantitative-benefit study rather than an observed outcome Strobbe et al. (2023). Reliability claims were the least supported among the major outcome domains: 56 reliability/mission-assurance/risk memberships were coded, but only 6 were moderate and none were strong.

2.6 | Verification and validation linkage

The detailed V&V synthesis coded study-level linkage categories from EPS-specific fields, pre-physical verification mechanisms, integration evidence, and test-readiness text. Categories are multi-label. Requirements-to-model traceability or verification matrices appeared in 33 studies; parametric, budget, or constraint checks in 29; test cases, procedures, or V&V activity packages in 29; EPS/power-specific pre-physical verification in 23; executable simulation or scenario analysis before build in 22; and external tool or data-flow coupling for V&V in 19. Seven studies had only generic or planned V&V linkage, six had no explicit EPS-focused pre-physical V&V, and four linked the model to physical, SIL/HIL, or integration-test readiness.

The scoping-review V&V categories were consistent with this pattern but less EPS-specific: 28 studies had V&V packages, reviews, plans, or procedures; 20 had parametric/resource checks; 16 linked executable simulation to the model; 13 linked requirements through satisfy/verify/trace relations; and 6 had no explicit model-linked V&V. The generated SLR coding sharpened this into pre-physical and EPS-focused categories.

Requirements traceability was usually represented through SysML satisfy/verify relations, verification matrices,

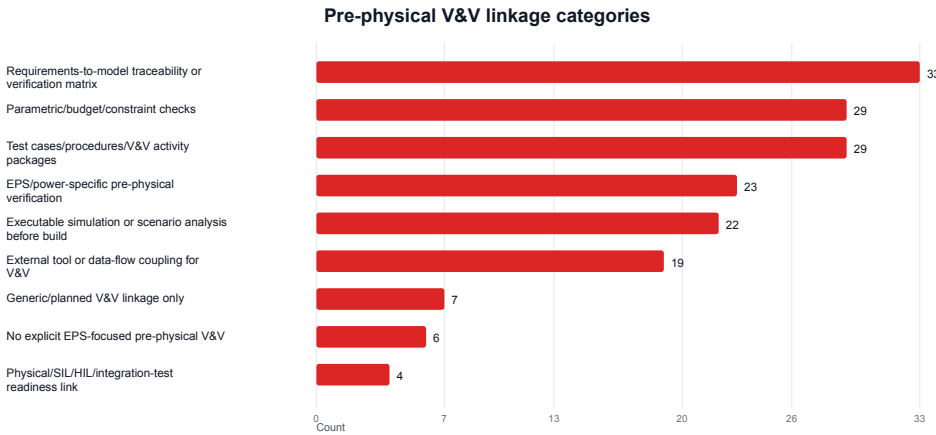


FIGURE 4 Pre-physical V&V linkage categories. Categories are multi-label by study.

TABLE 5 Pre-physical V&V linkage counts. Categories are multi-label by study.

V&V linkage category	n
Requirements-to-model traceability or verification matrix	33
Parametric/budget/constraint checks	29
Test cases/procedures/V&V activity packages	29
EPS/power-specific pre-physical verification	23
Executable simulation or scenario analysis before build	22
External tool or data-flow coupling for V&V	19
Generic/planned V&V linkage only	7
No explicit EPS-focused pre-physical V&V	6
Physical/SIL/HIL/integration-test readiness link	4

or tool-generated traceability views. In the EPS-specific case, requirement constraints and power-budget equations were extracted from Papyrus and evaluated by Java, producing respected/not-respected flags Strobbe et al. (2023). In academic nanosatellite project modeling, requirements were linked to "Verified By" test cases and behavior diagrams such as Bus System Activation and LEOP flow Hanafi et al. (2024). In the 3ColStar preliminary design, EPS continuous-power requirements were traced to physical components such as solar panels and batteries Buitrago-Leiva et al. (2024). In the NSS-1 SysML case, verify relationships connected mass and power requirements to constraint blocks, and positive margins were used as verification evidence Waseem and Sadiq (2018).

Parametric and executable V&V evidence was strongest when requirement checks were tied to simulation or budget outputs. RAX/CubeSat studies executed mission scenarios and trade studies involving energy collection, battery capacity, solar-panel sizing, data storage, and downlink Kaslow et al. (2014); Spangelo et al. (2013); Anderson et al. (2014). Later studies linked pre-physical V&V to open-source model-to-analysis pipelines Cordero et al. (2022); Luccisano et al. (2024); Campioli et al. (2025), PLM/Dymola requirements-in-the-loop simulation Rehman and Fortin (2024), and electrical-architecture energy-balance simulations before hardware development Boan (2025). Test-case or automated-test evidence was more limited and often not EPS-specific, as in the requirements modeling/mapping matrix for CubeSat tests Alanazi et al. (2019) and the EIRSAT-1 communications verification model coupled to Python and ground-station hardware Akarapu et al. (2023). The small count for physical/SIL/HIL/integration-test readiness ($n = 4$) shows that most V&V linkage stopped at pre-physical analysis, simulated verification, or planned test artifacts rather than demonstrated hardware or integration testing.

2.7 | Gap mapping

The extraction schema mapped each study to five gap categories adapted from the broader space-MBSE gap frame: practical implementation pathway; toolchain fragmentation/interoperability; lightweight/resource-constrained tailoring; governance/project deliverables/ROI/value; and NewSpace/agile/distributed/rapid iteration. Classifications were coded as addressed with evidence, addressed conceptually, mentioned only, or not addressed.

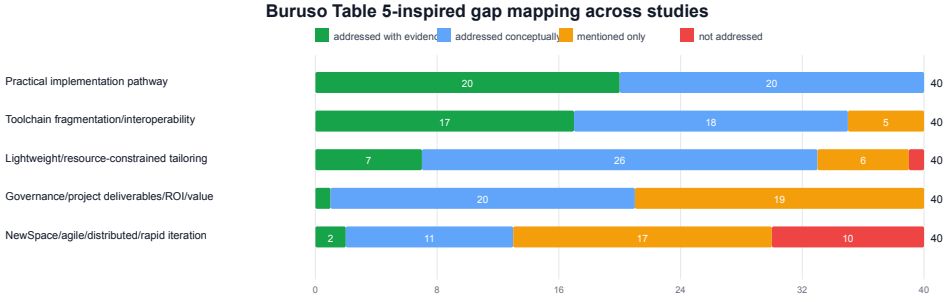


FIGURE 5 Gap-mapping counts for the extracted studies.

Implementation pathways were the most consistently addressed gap: all 40 studies addressed the pathway at least conceptually, split evenly between evidence-backed and conceptual classifications. Evidence-backed examples included concrete modeling/data-flow steps for EPS traceability Strobbe et al. (2023), concurrent electrical-architecture simulation workflow Boan (2025), and Capella/Arcadia plus agile preliminary-design execution Buitrago-Leiva et al. (2024). Toolchain fragmentation/interoperability was also frequently addressed, with 17 evidence-backed

TABLE 6 Gap mapping counts.

Gap	Evidence	Conceptual	Mentioned	Not addressed
Practical implementation pathway	20	20	0	0
Toolchain fragmentation/interoperability	17	18	5	0
Lightweight/resource-constrained tailoring	7	26	6	1
Governance/project deliverables/ROI/value	1	20	19	0
NewSpace/agile/distributed/rapid iteration	2	11	17	10

cases and 18 conceptual cases. Evidence-backed toolchain examples included executable RAX integrations Kaslow et al. (2014); Spangelo et al. (2013), Papyrus-to-Java EPS analysis Strobbe et al. (2023), open-source model-to-simulation coupling Campioli et al. (2025), and model-to-test/tool integration in EIRSAT-1 Akarapu et al. (2023).

Lightweight/resource-constrained tailoring was addressed with evidence in 7 studies, addressed conceptually in 26, mentioned only in 6, and not addressed in 1. Evidence-backed tailoring appeared in studies with rapid analytical design-space exploration Lowe and Macdonald (2014), academic project-management and design tailoring Hanafi et al. (2024), open/concurrent preliminary-design tooling Cordero et al. (2022), docs-as-code for a university CubeSat Chiu et al. (2023), causal MBSE user evaluation Peters et al. (2022), low-cost tradespace modeling Nardi et al. (2025), and agile distributed preliminary design Buitrago-Leiva et al. (2024).

Governance, project deliverables, ROI, and value linkage was the weakest gap category in the corpus: only one study was coded as addressed with evidence, 20 were conceptual, and 19 mentioned the issue only. The evidence-backed case was the 3ColStar study, which reported Capella/Arcadia governance, weekly sprints, requirement/design traceability, and a four-month preliminary design cycle Buitrago-Leiva et al. (2024). Other studies often reported potentially relevant artifacts, such as PBS/WBS/CBS derivation Hanafi et al. (2024) or cost/schedule estimates in concurrent design Bella et al. (2024), but the coded gap evidence did not rise to a broader governance/ROI demonstration. NewSpace/agile/distributed/rapid-iteration evidence was also limited, with 2 evidence-backed studies, 11 conceptual, 17 mentioned-only, and 10 not addressed.

3 | DISCUSSION

3.1 | Relationship to the broader space-MBSE review

Buruso et al. provide the appropriate background for this review because their comprehensive space-MBSE synthesis already covers the broad distribution of MBSE methods, modeling languages, tools, lifecycle phases, and enabler classes across space applications Buruso et al. (2026). The present review therefore does not treat the recurrence of SysML, Cameo/MagicDraw, Capella/Arcadia, PLM tools, simulation environments, or integrated digital-engineering toolchains as a novel finding by itself. Instead, those enablers are interpreted only where they affect nanosatellite and EPS-relevant value evidence, implementation burden, or model-linked V&V.

The strongest connection to Buruso et al. is their discussion of value-driven implementation and their Table 5 research gaps. They identify persistent gaps in practical implementation pathways, fragmented toolchains and lifecycle interoperability, resource-mismatched frameworks for small missions, weak linkage between MBSE artifacts, governance and ROI, and misalignment with NewSpace-style development needs Buruso et al. (2026). Our results support those gaps in a narrower population and make their nanosatellite consequences more explicit. In the extracted corpus, practical implementation pathways and toolchain interoperability are frequently addressed, with evidence-backed classifications in 20 and 17 studies respectively, but governance/ROI/value and NewSpace/agile/distributed development are rarely addressed with evidence (1 and 2 studies respectively; Table 6). Thus, the nanosatellite literature is

not simply missing MBSE awareness; it is missing durable evidence that the models change project outcomes, review decisions, or downstream verification practice.

This narrower focus also exposes an EPS-specific asymmetry that is hidden in broad space-MBSE taxonomies. Many studies include EPS or power concerns inside whole-CubeSat models, but only one extracted study is EPS-specific or EPS-first Strobbe et al. (2023). Most examples of power modeling are embedded in system-level trade studies, reference-model work, or preliminary design workflows such as RAX energy/data simulation Kaslow et al. (2014); Spangelo et al. (2013), SysML power-budget margins Waseem and Sadiq (2018), Capella/Arcadia preliminary EPS allocation Buitrago-Leiva et al. (2024), PLM/Dymola battery simulation Rehman and Fortin (2024), and electrical-architecture concurrent engineering Boan (2025). This extends Buruso et al.'s Table 5 concern about implementation pathways: for nanosatellites, the missing pathway is not only "how to adopt MBSE," but how to maintain an EPS/system thread from mission objectives to power requirements, architectural allocation, analysis, verification procedure, and test evidence without imposing enterprise-scale overhead.

3.2 | Value-driven MBSE remains under-evidenced for nanosatellites

The review supports Buruso et al.'s claim that MBSE value must be tied to explicit objectives, model usage, governance, and evaluation criteria Buruso et al. (2026). The nanosatellite corpus contains many value claims, but the evidence is generally modest. All 40 studies were only partially relevant to RQ2, and among 294 extracted value-claim records no claim was coded as strong evidence; 97 were moderate, 152 weak, and 45 unsupported (Table 1). The most common domains were decision support, traceability, reproducibility/reuse, interoperability, and pre-physical V&V, but the reliability/mission-assurance domain was comparatively weak, with only 6 moderate memberships and no strong evidence (Table 4).

Moderate evidence usually came from executable demonstrations or traceable artifacts rather than from measured project outcomes. For example, the RAX studies show how an integrated SysML–ModelCenter–STK–MATLAB workflow can simulate energy and data interactions, support solar-panel and battery-capacity trades, and check mission requirements Kaslow et al. (2014); Spangelo et al. (2013). The open-source Papyrus power-subsystem study demonstrates parameter-impact traceability and requirement-constraint flags from model data Strobbe et al. (2023). PLM/Dymola requirements-in-the-loop simulation reports battery state-of-charge behavior against an SoC requirement Rehman and Fortin (2024), and the electrical-architecture workflow reports a worst-case battery state-of-charge margin for survival/no-illumination conditions Boan (2025). These are useful forms of engineering evidence, but they are mostly prototype, single-case, or pre-physical demonstrations.

Evidence becomes thinner when claims move from technical feasibility to project value. Reported schedule or cost benefits exist, such as a one-month versus three-month feasibility-study comparison and lower estimated mission cost in a concurrent-design case Bella et al. (2024), or a four-month preliminary-design cycle with Arcadia and weekly sprints for 3ColStar Buitrago-Leiva et al. (2024). However, such claims are case-specific and rarely supported by controlled baselines, repeated use, independent stakeholder assessment, or lifecycle follow-through. Similarly, reference models and templates promise reuse and consistency Kaslow et al. (2020); Kaslow and Madni (2017); Kaslow et al. (2019), while docs-as-code and open-source workflows improve accessibility and reproducibility Chiu et al. (2023); Luccisano et al. (2024), but the corpus seldom demonstrates reuse across independent missions or quantifies the effort required to maintain the model.

The implication is that nanosatellite MBSE remains more artifact-driven than value-driven. Studies frequently show that a model can contain requirements, architectures, budgets, interfaces, simulations, or generated documents. Fewer studies define the mission/project objective that justifies the modeling effort, the metric used to evaluate suc-

cess, the baseline practice being improved, or the review gate at which the model changes a decision. This is especially important for EPS: power budgets, SoC traces, and margin checks are often convincing local analyses, but they are not always connected to explicit mission assurance objectives such as survival probability, operational availability, fault tolerance, or verification closure. In this respect, our synthesis directly supports Buruso et al.'s Table 5 gap on weak integration between MBSE artifacts, governance, ROI, and value realization, and shows that the gap persists even in a domain where resource constraints should make value justification central Buruso et al. (2026).

3.3 | V&V linkage is present but rarely continuous

The corpus does not show an absence of V&V linkage; rather, it shows many partial linkages that rarely form a continuous model-to-test thread. Requirements-to-model traceability or verification matrices appeared in 33 studies, parametric/budget/constraint checks in 29, test cases/procedures/V&V activity packages in 29, EPS/power-specific pre-physical verification in 23, executable simulation or scenario analysis in 22, and external tool or data-flow coupling in 19 (Table 5). These counts indicate that nanosatellite MBSE studies often recognize verification as a modeling concern. They also align with Buruso et al.'s observation that later lifecycle testing, integration, and verification have poorer visibility in the wider space-MBSE literature Buruso et al. (2026).

The most common V&V pattern is requirements traceability plus a local analytical check. The EPS-specific Papyrus study extracts requirement constraints and power-budget parameters to generate respected/not-respected flags Strobbe et al. (2023). The academic nanosatellite framework links requirements to "Verified By" test cases and behavior diagrams such as bus activation and LEOP flow Hanafi et al. (2024). The 3ColStar study links continuous-power requirements to EPS components and mode-level power analyses Buitrago-Leiva et al. (2024). The NSS-1 SysML case uses verify relationships and parametric constraints to compute positive mass and power margins Waseem and Sadiq (2018). These examples show useful pre-physical verification, but the verification often stops at traceability, calculation, or planned test artifacts.

Executable simulation provides a second important but still partial pattern. RAX-style workflows connect SysML behavior and parametrics to STK, MATLAB, and ModelCenter to check energy, data, and mission-scenario behavior Kaslow et al. (2014); Spangelo et al. (2013). Other studies link model data to power/data-budget simulation or mission-mode simulation Cordero et al. (2022); Luccisano et al. (2024); de Almeida et al. (2021); Campioli et al. (2025), PLM/Dymola battery behavior Rehman and Fortin (2024), and electrical-architecture energy-balance simulation Boan (2025). These studies strengthen pre-build confidence and support design trades, but they do not generally show continuity from the same model elements to formal test procedures, hardware-in-the-loop execution, integration-test data, anomaly handling, or acceptance evidence.

The few test-oriented examples reinforce this distinction. Requirements Modeling Language and RMM work links requirements, code, tests, and defects for CubeSat software-oriented examples Alanazi et al. (2019). EIRSAT-1 demonstrates a communications-subsystem verification model coupled through Python to ground-station hardware and requirements-compliance updates Akarapu et al. (2023). These are among the clearest model-to-test patterns in the corpus, but they are not EPS demonstrations. For EPS, the literature more often contains power budgets, margin calculations, and simulations than a continuous thread from EPS requirement to model allocation, analysis case, test case, test procedure, measured test data, and verification closure. This is the central V&V gap for nanosatellite MBSE: linkage exists, but it is fragmented across requirements traceability, parametric checks, simulation, and planned V&V packages.

3.4 | Implications for lightweight nanosatellite MBSE practice

For resource-constrained CubeSat and nanosatellite teams, the practical implication is not to adopt more MBSE enablers, but to adopt a thinner value-and-verification thread. Buruso et al.'s Table 5 calls for modular, scalable MBSE frameworks and adoption playbooks for lightweight missions Buruso et al. (2026). Our synthesis suggests that such playbooks should start by defining the value hypothesis for the model: which decision, review, risk, or verification activity the model is expected to improve; what artifact will be produced; what metric or comparator will indicate success; and when the modeling effort should stop. Without these criteria, teams can accumulate model artifacts while still lacking evidence that MBSE improved reliability, reproducibility, integrability, or schedule performance.

A lightweight nanosatellite playbook should make EPS a first-class model thread rather than a subsystem table appended to the architecture. At minimum, the thread should connect mission modes and operational scenarios to EPS requirements, power-generation/storage/distribution architecture, assumptions, margins, budget equations, executable checks, verification methods, and test cases. Studies already provide pieces of this thread: open-source Papyrus/SysML traceability for a power subsystem Strobbe et al. (2023), model-derived power and data budgets Luccisano et al. (2024), Capella/PVMT simulation coupling Campioli et al. (2025), PLM/Dymola requirements-in-the-loop behavior Rehman and Fortin (2024), and concurrent electrical-architecture simulation Boan (2025). The missing practice guidance is how a small team should select a minimal subset of these mechanisms and maintain it through review and test readiness.

Tool choices should be evaluated against adoption burden as well as analytical capability. Enterprise or integrated environments can provide powerful lifecycle continuity, as shown by PLM/Dymola and concurrent-engineering workflows Rehman and Fortin (2024); Boan (2025), but they may reproduce the resource-mismatch problem noted by Buruso et al. Buruso et al. (2026). Conversely, open-source, plain-text, or docs-as-code approaches can reduce licensing and training barriers while supporting traceability and generated deliverables Strobbe et al. (2023); Luccisano et al. (2024); Chiu et al. (2023). The OPAL user study suggests that low-training conceptual-design support can be valuable for student teams Peters et al. (2022), but it also shows that usability is not equivalent to verification capability. Lightweight practice therefore needs an evidence ladder: accessibility and usability are entry criteria, while value realization requires traceable decisions, executed checks, and eventually test closure.

The strongest practice pattern in the corpus combines bounded scope, explicit roles, and recurring review cadence. The 3ColStar case links Capella/Arcadia modeling with weekly sprints, permissions, subsystem responsibilities, traceability, and preliminary-design outputs Buitrago-Leiva et al. (2024). Academic project-management frameworks similarly connect model artifacts to PBS/WBS/CBS and review gates, although they do not quantify ROI Hanafi et al. (2024). Such examples suggest that adoption playbooks should include not only modeling diagrams and tool scripts, but also governance: who owns requirements, who approves assumption changes, how power-budget changes propagate, what evidence is required at PDR/CDR-like gates, and which artifacts are frozen or regenerated for test planning.

3.5 | Threats to validity and limitations of this review

Several threats limit the strength and generality of this discussion. The search scope was limited primarily to Scopus and IEEE Xplore, with Google Scholar used only for exact-title fallback during targeted recovery. Grey literature, internal mission documentation, tool repositories without papers, and industrial reports may contain stronger evidence for MBSE value or model-linked V&V than the accessible academic corpus. Four retained reports were not retrieved or remained inaccessible and were excluded from extraction, so the synthesis reflects 40 accessible studies rather than the full set of potentially relevant records.

Extraction and synthesis also have reliability limits. The review used a fixed protocol, standardized extraction schema, direct evidence quotes, quality appraisal, and extraction-confidence ratings, but no independent inter-rater reliability exercise is documented. Claim-evidence confidence, EPS relevance, V&V linkage, and Table 5-inspired gap mappings therefore include reviewer judgment. Deterministic category rules helped make counts reproducible, but they can simplify nuanced paper content, especially where authors use broad terms such as validation, verification, digital engineering, traceability, or model integration differently.

The evidence base itself is immature. The scoping limitations coded 31 studies as prototypes, incomplete demonstrations, or future work; 24 as limited-scope, single-case, or partial subsystem/mode studies; 24 as lacking quantitative evaluation, having weak validation, or omitting a threats discussion; 18 with data/model-quality/source/assumption limits; and 16 with tool-dependency, interoperability, licensing, or learning-curve limits. Repeated publications in the CubeSat System Reference Model lineage also mean that publication count should not be interpreted as independent replication. Because only one extracted study is EPS-specific, EPS conclusions necessarily rely on a mixture of whole-nanosatellite, EPS-adjacent, and power-budget/electrical-architecture studies.

Use of Buruso et al.'s Table 5 gaps is another limitation. The gaps were used as a positioning and sensitizing frame, not as an independent coding validation instrument Buruso et al. (2026). Mapping nanosatellite studies onto those broad gaps is useful for comparison, but the resulting categories are coarse and should be interpreted as evidence-strength descriptors rather than definitive classifications of each study.

References

Sai Krishna Reddy Akarapu, Bas Stijnen, Caimin McKenna, David J. McKeown, D. Murphy, Jack Reilly, Joseph W. Thompson, M. Doyle, R. Wall, S. McBreen, and L. Hanlon. Development of a Nanosatellite System Modeling Architecture for EIRSAT-1. 2023. https://doi.org/10.1109/smc-it56444.2023.00022.

Abdulaziz Alanazi, Andrew B. Jones, and Jeremy Straub. Requirements Modeling Language and Automated Testing for CubeSats. In *IEEE AUTOTESTCON 2019*. IEEE, 2019. https://doi.org/10.1109/autotestcon43700.2019.8961058.

L. Anderson, B. Cole, Rose Yntema, Manas Bajaj, Sara C. Spangelo, D. Kaslow, C. Lowe, Eric V. Sudano, M. Boghosian, R. Reil, S. Asundi, and Sanford A. Friedenthal. Enterprise Modeling for CubeSats. In *2014 IEEE Aerospace Conference*. IEEE, 2014. https://doi.org/10.1109/aero.2014.6836343.

Emanuela La Bella, Serena Campioli, Luisa Iossa, Sabrina Corpino, and Fabrizio Stesina. Concurrent Model-Based Approach for CubeSat Mission Design. In *75th International Astronautical Congress (IAC), Milan, Italy, 14-18 October 2024*. International Astronautical Federation (IAF), 2024. https://doi.org/10.52202/078374-0035.

Pierre Boan. CONCURRENT ENGINEERING PROCESS FOR ELECTRICAL ARCHITECTURE OF NANOSATELLITES PRELIMINARY PHASES, 2025.

Jeimmy Nataly Buitrago-Leiva, Juan José Mejía, Juan Francisco Puerta-Ibarra, Ignacio Francisco Acero-Niño, Andrés Felipe Guarnizo-Saavedra, Julian Rodriguez-Ferreira, Leandro Rojas-Rodriguez, Francisco Luis Hernández-Torres, Cristian Esteban Arango-Cotacio, Jorge Enrique Salazar-Morales, Miguel Angel Herrera-Cruz, Mario Linares-Vásquez, Jose Fernando Jiménez-Vargas, Jorge Enríque Espíndola-Díaz, Óscar Javier Montañez-Sogamoso, and Adriano Camps. Preliminary Design of Satellite Systems through the Integration of Model-Based System Engineering and Agile Methodologies: Application to the 3ColStar Mission. *Aerospace*, 2024. https://doi.org/10.3390/aerospace11090758.

Rehobot Bekele Buruso, Sara Himmiche, Taiwo Tejumola, Yeshirun Alemayehu, and Bertrand Rose. Model-Based Systems Engineering in Space Applications: A Comprehensive Literature Review. *Systems Engineering*, 0:1–43, 2026. https://doi.org/10.1002/sys.70051.

Serena Campioli, Giacomo Luccisano, Davide Ferretto, and Fabrizio Stesina. Towards a Unified Modeling and Simulation Framework for Space Systems: Integrating Model-Based Systems Engineering with Open Source Multi-Domain Simulation Environments. *Aerospace*, 2025. https://doi.org/10.3390/aerospace12080745.

Kevin Chiu, Sean Marquez, and Sharanabasaweshwara A. Asundi. Model Based Systems Engineering with a Docs-as-Code Approach for the SeaLion CubeSat Project. *Systems*, 2023. https://doi.org/10.3390/systems11070320.

Sophia Salas Cordero, Thibault Gateau, and Rob Vingerhoeds. MBSE Challenges in the Concurrent Preliminary Design of CubeSats: Nanospace Study. IEEE, 2022. https://doi.org/10.1109/syscon53536.2022.9773818.

Danilo Pallamin de Almeida, Bence Graics, Ronan Arraes Jardim Chagas, Fabiano Luis de Sousa, and Fátima Mattiello-Francisco. Towards Simulation of CubeSat Operational Scenarios under a Cyber-Physical Systems View. 2021. https://doi.org/10.1109/ladc53747.2021.9672594.

Ahmed Hanafi, Zakaria Moutakki, Mohamed Karim, and Tajjeeddine Rachidi. Effective Model-Based Systems Engineering framework for academic nanosatellite project management and design. 2024. https://doi.org/10.1007/s12567-024-00542-2.

D. Kaslow, P. T. Cahill, and B. Ayres. Development and Application of the CubeSat System Reference Model. 2020. https://doi.org/10.1109/aero47225.2020.9172714.

David Kaslow and Azad M. Madni. Validation and Verification of MBSE-compliant CubeSat Reference Model. In *15th Annual Conference on Systems Engineering Research: Disciplinary Convergence: Implications for Systems Engineering Research*. Springer, 2017. https://doi.org/10.1007/978-3-319-62217-0_27.

David Kaslow, Grant Soremekun, Hongman Kim, and Sara Spangelo. Integrated Model-Based Systems Engineering (MBSE) Applied to the Simulation of a CubeSat Mission. In *2014 IEEE Aerospace Conference*. IEEE, 2014. https://doi.org/10.1109/aero.2014.6836317.

David Kaslow, Philip T. Cahill, and Rodger Frank. Developing a CubeSat System MBSE Reference Model – Interim Status #5. In *Small Satellite Conference*, 2019.

C. J. Lowe and M. Macdonald. Rapid Model-based Inter-disciplinary Design of a CubeSat Mission. *Acta Astronautica*, 2014. https://doi.org/10.1016/j.actaastro.2014.10.002.

Giacomo Luccisano, Sophia Salas Cordero, Thibault Gateau, and Nicole Viola. Open-Source Data Formalization through Model-Based Systems Engineering for Concurrent Preliminary Design of CubeSats. *Aerospace, 11(9), Article 702*, 2024. https://doi.org/10.3390/aerospace11090702.

Lorenzo Nardi, Stefano Carletta, Parsa Abbasrezaee, Giovanni Palmerini, Nicola Lovecchio, Nunzio Burgio, Alfonso Santagata, Massimo Frullini, Donato Calabria, Massimo Guardigli, Elisa Michelini, Maria Maddalena Calabretta, Martina Zangheri, Elisa Lazzarini, Andrea Pace, Marco Montalti, Dario Mordini, Liyana Popova, Saverio Citraro, Daniela Billi, Fabio Lorenzini, Alessandro Donati, Mara Mirasoli, and Augusto Nascetti. Integrating Model-Based Systems Engineering into CubeSat Development: A Case Study of the BOREALIS Mission. *Aerospace, 12(3), Article 256*, 2025. https://doi.org/10.3390/aerospace12030256.

Stephen Peters, Clément Fortin, and Grant McSorley. Investigating the Effectiveness of a Causal MBSE Software Tool in Modelling CubeSat Conceptual Design Data. Springer (not explicitly shown in markdown), 2022. https://doi.org/10.1007/978-3-030-94399-8_10.

Mubeen Ur Rehman and Clement Fortin. An Implementation of Integrated Approach in Product Life-cycle Management Tool to Ensure Requirements-In-Loop During Complex Product Development: A Cubesat Case Study. Springer, 2024. https://doi.org/10.1007/978-3-031-62578-7_21.

Sara C. Spangelo, James Cutler, L. Anderson, E. Fosse, Leo Y. Cheng, Rose Yntema, Manas Bajaj, Chris Delp, Bjorn Cole, Grant Soremekum, and D. Kaslow. Model based systems engineering (MBSE) applied to Radio Aurora Explorer (RAX) CubeSat mission operational scenarios. In *IEEE Aerospace Conference*. IEEE, 2013. https://doi.org/10.1109/aero.2013.6496894.

Charlotte Strobbe, Sophia Salas Cordero, and Rob Vingerhoeds. Open-source CubeSat MBSE Approach to Address Traceability: Power Subsystem. In *IEEE Systems Conference (SysCon), inferred from DOI string; venue not printed in Markdown*. IEEE, inferred from DOI prefix 10.1109, 2023. https://doi.org/10.1109/syscon53073.2023.10131054.

Muhammad Waseem and Muhammad Usman Sadiq. Application of model-based systems engineering in small satellite conceptual design-A SysML approach, 2018.